Early Warning of Pitcher Arm Injuries from Pitch-Tracking Time Series

MLB Statcast, 2021–2026 regular seasons

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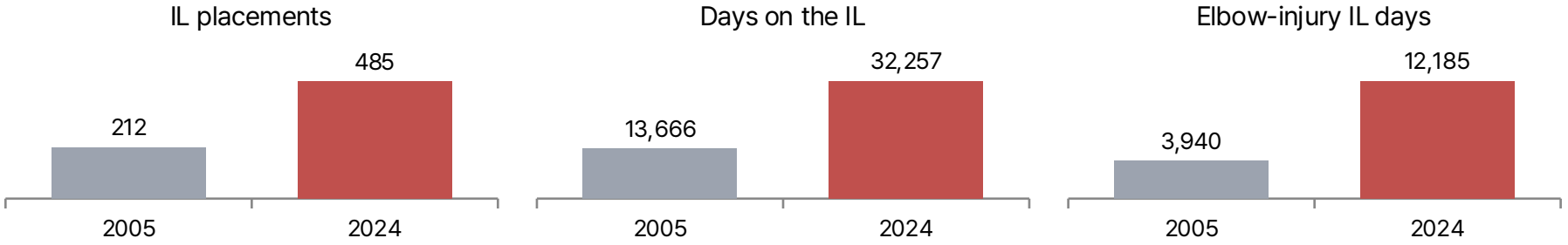
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Background

Problem statement: what and why

Goal Detect departures from a pitcher's own normal in per-appearance velocity, velocity variability, and release point, and raise a warning before an arm-injury IL placement



MLB pitcher injuries, 2005 vs 2024 (AP)

- IL placements **212 → 485**, IL days **13,666 → 32,257**; elbow-injury IL days roughly **tripled** (3,940 → 12,185)
- Shoulder and elbow injuries, 2015–2025: **1,122** pitchers, **2,365** IL placements, about **$3.33B** in lost salary (Dillon et al., OJSM)
- Once a pitcher is on the IL, the loss has already happened. The real question is not who is at risk but **when to rest him**
- Catching early signs lets teams adjust rest days, manage workload, or give rest

Background

Why this is a time-series problem

Autocorrelation

- The signal is defined only as a deviation from the pitcher's own past
- Shuffle the order of appearances and the signal disappears
- So we can forecast the next value from the past and monitor the residual

Trend

- The 'velocity revolution' keeps raising league four-seam velocity
- 93.69 → 94.77 mph (2021→2026): +1.08, up every single season
- A component that must be removed before judging a pitcher's anomaly

Seasonality

- Velocity is higher later in the season than early on
- 2025: April 94.18 → June 94.61 mph, then flat; 2024 has the same shape (94.04 → 94.40)
- Injuries themselves also tend to cluster in certain parts of the season

Exogenous variables

- Rest days (days since the last appearance)
- Pitches thrown in the last 7 / 14 / 28 days, ACWR (acute : chronic workload)
- Separate velocity changes explained by rest and workload from injury signals

Related work

(1) Velocity–injury link · normal vs. pre-injury

Study	Data · method	Key result	Limitation (vs. ours)
Topic 1. Studies supporting the premise that velocity is linked to injury			
Prodromo et al. (2016)	UCLR pitchers vs. age-matched controls (2002–2015) Case-control, conditional logistic regression	Higher velocity in the season before surgery; mean fastball velocity was significant	Compares pitchers with season averages → cannot tell when change happens
Topic 2. Studies comparing a pitcher's normal with the pre-injury period			
Mastroianni et al. (2026)	78 UCL-surgery pitchers vs. 156 matched controls Within-pitcher change	Gradual velocity drop over the last 5 appearances (P = .019); larger horizontal release variability (P = .005)	Retrospective; no AUC or lead time reported
Ozaki (2026, MS thesis)	46 UCLR pitchers (2016–2024) Autoencoder anomaly detection	Anomaly grew 2.7× closer to injury; code and 45 matched pairs released	Healthy pitchers also flagged (73% vs. 66%); time order ignored on purpose

Related work

(2) Time-series methods · exogenous variables

Study	Data · method	Key result	Limitation (vs. ours)
Topic 3. Studies using time-series methods			
Mayo et al. (2021)	Last 15 games before UCLR, 223 pitchers Mann-Kendall trend test	Four-seam velocity fell ~0.1 mph per game (tau = −0.657, P < .001); spin fell ~5 rpm per game	No healthy controls; no alert rule
Bright et al. (2026)	3D pose from broadcast video Screened 7,348 pitchers	AUC 0.811 for Tommy John surgery, 0.825 for severe arm injury	Video estimates; point-in-time screening, not a per-pitcher time-series alert
Topic 4. Studies on exogenous variables			
Slowik et al. (2021)	17 college pitchers ACWR with a ROC-chosen cutoff	ACWR change beyond ±33% → 8.3× next-week injury risk	Small sample (95% CI 1.8–54.1); 3-week chronic window

Related work

Limitations of previous studies

Limitation 1

Injured vs. healthy pitchers compared with season averages

What is needed

Read change against each pitcher's own time series

Limitation 2

Trend tests only: no alert rule, no false-alarm count

What is needed

An alert system judged by lead time and false alarms

Limitation 3

Appearances treated as independent or averaged over a season (autocorrelation ignored)

What is needed

A time-series model that forecasts the next appearance from past ones and monitors the residual

Related work

Novelty of our approach

Core idea a per-pitcher time-series baseline; monitor the gap (residual) between actual velocity and the velocity forecast from his own past

1. Multivariate time series on open data

Performance velocity · variability · release

Workload rest days · pitch counts · ACWR

Uses only public Statcast and MLB records → reproducible results

2. Evaluating a real alert system, not just a test

How early? days before the IL placement

How often wrong? false-alarm rate

None of the studies we reviewed reports both lead time and false-alarm rate

Data

Dataset: MLB Statcast (2021–2026)

- **Sources**
  - Pitch data: Baseball Savant **Statcast**, pitch by pitch
  - Injury labels: **MLB official API transactions** (IL placement date, body part)
  - Temperature: MLB API game weather, filled with Open-Meteo if needed
- **Data info**

Item	Description
Period	2021–2026 regular seasons (6), after the 2017 velocity redefinition and 2020 Hawk-Eye switch
Size	~4.38M pitches · 855–909 pitchers per season · 3,340 IL placements (1,683 arm-related)
Frequency	Pitches aggregated per pitcher × appearance (irregular spacing)
Key variables	Velocity, release side / height, extension, spin rate, pitch type, pitcher, date, game ID
Exogenous	Rest days, 7 / 14 / 28-day pitch counts, ACWR

- **Suitability**: a long, open, pitch-level series linked to real injury labels, so warnings can be checked against what actually happened

Data

Unit of analysis and variables

Group	Variable	Definition
Performance	**Velocity**	Mean & max fastball velocity per appearance
	Velocity variability	SD of fastball velocity per appearance
	Release point	Mean & SD of horizontal / vertical release
	Auxiliary	Spin rate, extension, in-game velocity decay slope
Workload	**Rest days**	Days since the last appearance
	Recent load	Cumulative pitches over 7 / 14 / 28 days
	ACWR	7-day ÷ 28-day average load

- **Unit**: pitcher × appearance; time axis = appearance index, rest days as a separate variable
- **Injury event**: IL placement for an arm-related reason
- IL records are used **only for evaluation**, never for training

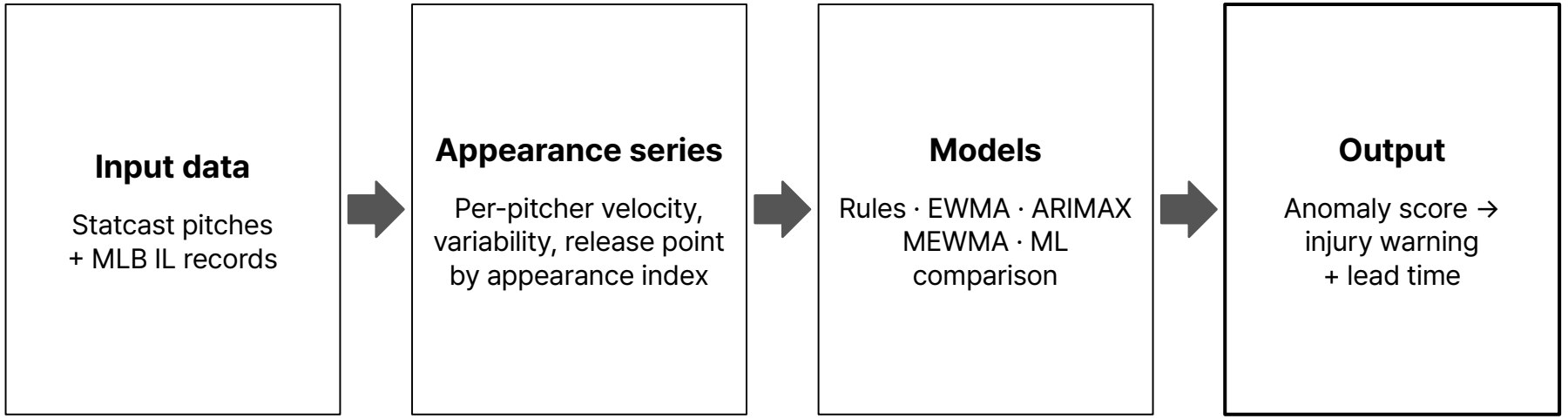
Method

Research questions

RQ1	How early and how accurately can the residual (actual – velocity forecast from the pitcher's own past) detect an arm-injury IL placement?
RQ2	In the last few appearances before injury, do velocity, its variability, and release point move away from the pitcher's own normal?
RQ3	Do workload indicators such as rest days and 7 / 14 / 28-day pitch counts improve alert performance?
RQ4	Does the strength of the warning signal differ between hard throwers and other pitchers?

Method

Framework



Alert success alert within the last 3 appearances (~21 days) before IL; 2 and 5 also tested, fixed on validation

Method

Data split · models · evaluation

Time-ordered split	Train 2021–2024	Valid. 2025 (threshold)	Test 2026

Models

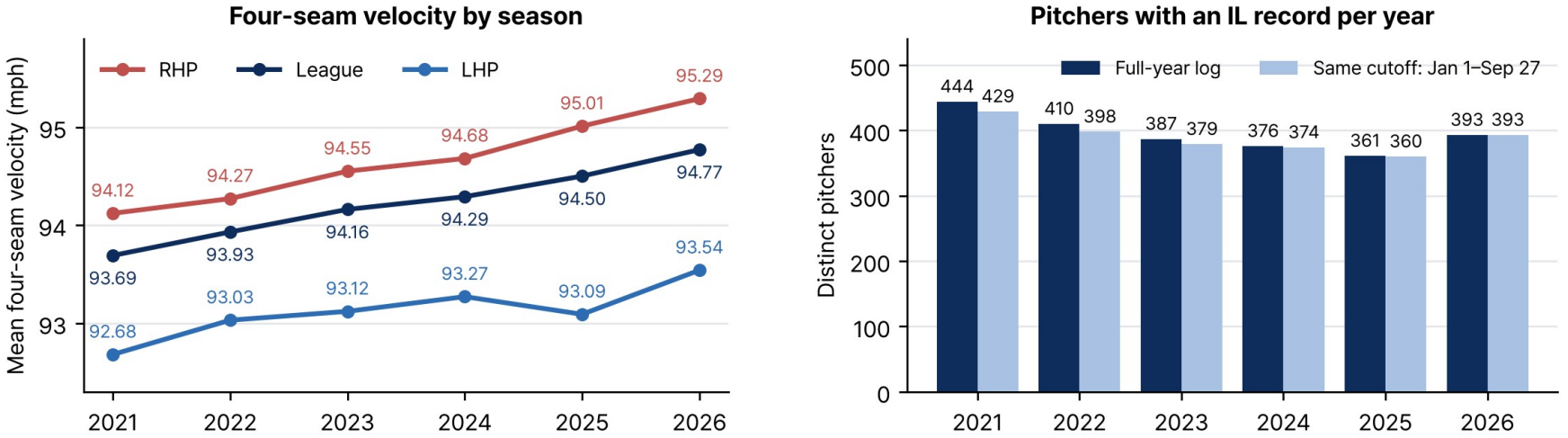
Type	Model	Description
Baselines	1. Random score	Chance level
	2. Simple rule	Alert when velocity drops X mph below normal
	3. Workload rule	Alert when ACWR moves beyond ±33%
Time series	4. EWMA	Cumulative detection of a gradual standardized drop
	5. ARIMAX	Residual vs. expected velocity given rest days and pitch counts
Extra check	6. ML methods	Compared on the same terms with methods that ignore time order
Extension	7. MEWMA	Monitors velocity, variability, release together (multivariate)

Evaluation metrics

Question	Metric
Accuracy	Recall and precision with ≤ 1 false alarm per pitcher-season (main: F1)
False alarms	False alarms per pitcher-season, alert rate in controls
Lead time	Days from first alert to IL placement (median, distribution)
Overall	Appearance-level AUPRC
Reference	Velocity forecast error: MAE, RMSE

Exploratory Data Analysis

(1) League velocity trend and IL volume

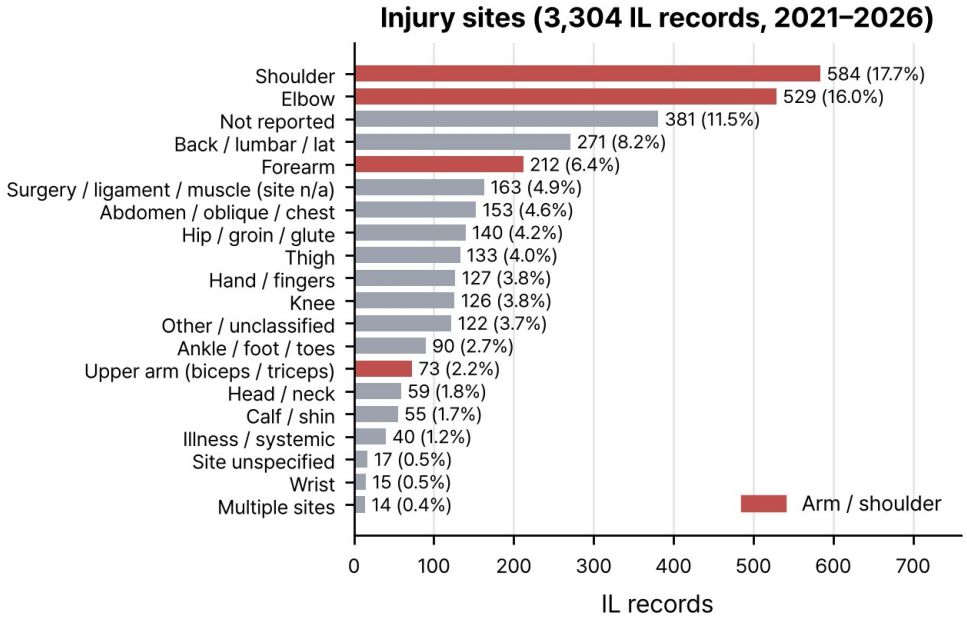


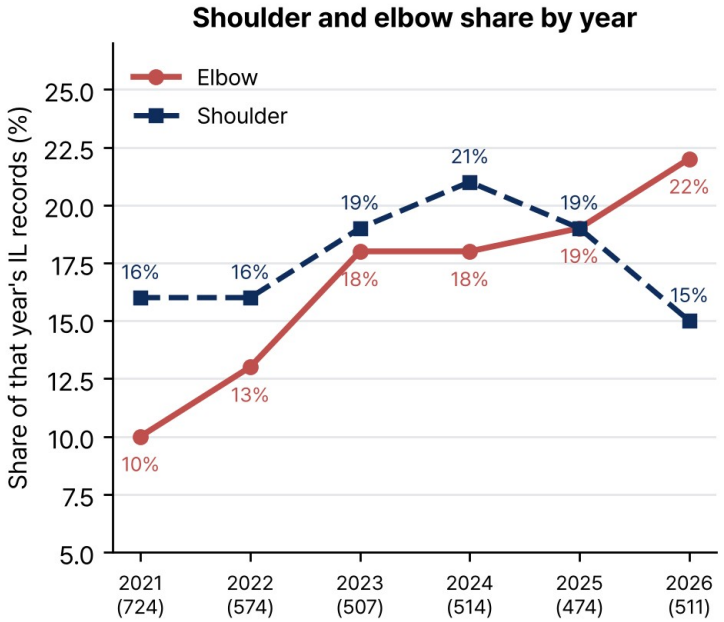
Four-seam velocity by season (pitch-weighted, left) and distinct pitchers with an IL record (right), 2021–2026

- League four-seam velocity rose six seasons in a row: **93.69 → 94.77 mph** (+1.08) → we need a detrended, per-pitcher baseline
- **360–444** pitchers per year had an IL record; the 2026 log ends Sep 27, so we also compare on the same cutoff (volume only, not an injury rate)

Exploratory Data Analysis

(2) Injury sites

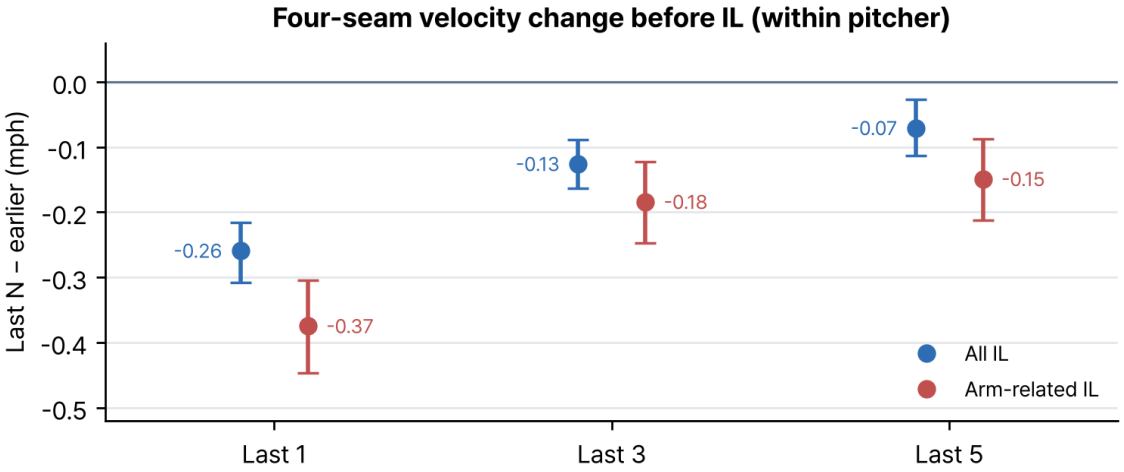




- Shoulder 17.7%, elbow 16.0%, forearm 6.4%: arm sites lead (denominator: IL records, not pitchers)
- Elbow share rose from **10% (2021) to 22% (2026)** → arm-related IL is our main event

Exploratory Data Analysis

(3) Velocity right before IL placement



IL events	All IL	Arm-related
Last 1	1,479	663
Last 3	1,066	460
Last 5	786	329

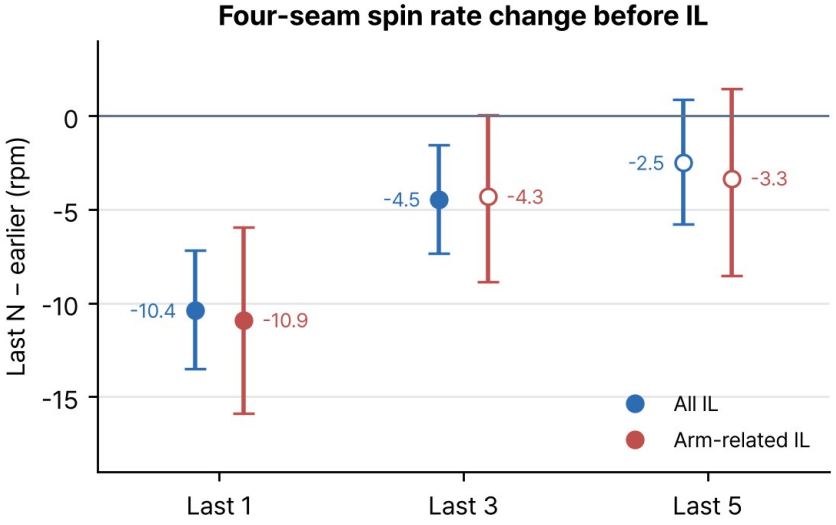
Within the same pitcher's current IL segment: mean of last N appearances – mean of earlier ones (dot: difference, line: 95% CI, 2,000 pitcher-cluster bootstraps)

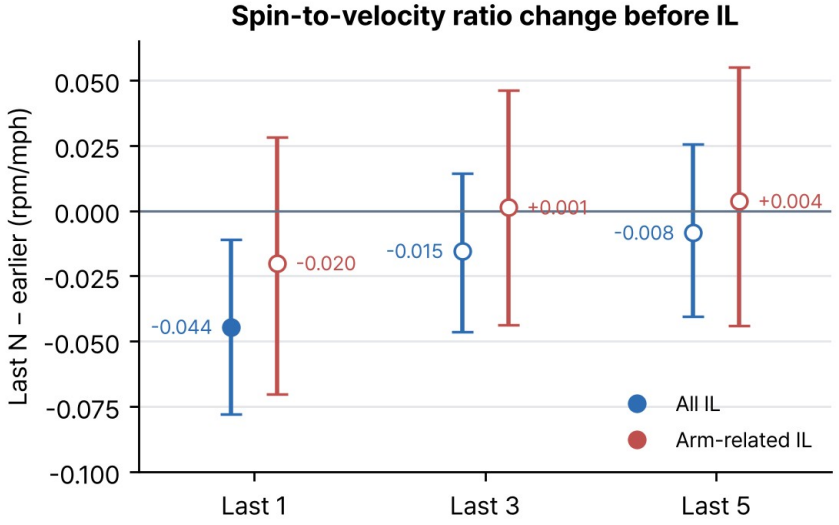
- This is a **within-pitcher comparison**, not a comparison with healthy pitchers
- The drop is largest in the last appearance (all −0.26, arm −0.37 mph) and shrinks as N grows; arm-related IL drops more in every window

➡ The drop is concentrated right before IL → a short alert window (last 3, checked with 2 and 5)

Exploratory Data Analysis

(4) Spin rate and spin-to-velocity ratio before IL



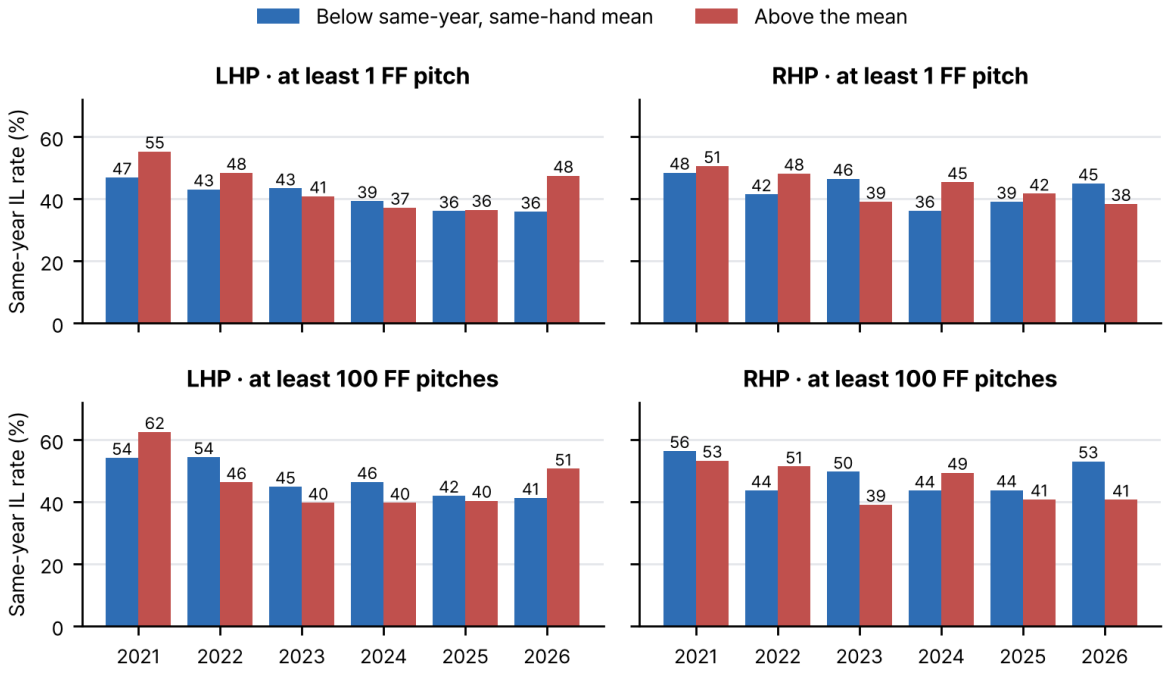


Last N – earlier appearances (within pitcher). Filled dot = 95% CI excludes 0; open dot = includes 0

- Spin rate: about **−10 rpm** in the last appearance (all −10.4, arm −10.9); over the last 5 the 95% CI includes 0
- Spin-to-velocity ratio: only the last appearance for all IL is below 0 (−0.045); for arm-related IL every window includes 0

Exploratory Data Analysis

(5) Same-year IL rate by velocity group



Same-year IL rate for pitchers above (red) and below (blue) the same-year, same-hand mean

- Compares groups of pitchers: faster or slower than the mean for that year and throwing hand
- The faster group is higher in **12 of 24** year × group cells; no consistent direction

Thank you

Q & A

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